

Robot URDF Model

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1. Course Content

Basic: Learn how to view the robot's model file and use the GUI debugging tool window to check the movement of the model's joints by sliding the progress bar.

Advanced: Understand the composition of a URDF description file.

2. Preparations

2.1 Content Description

For Raspberry Pi boards, you need to open a terminal on the host machine, enter the command to get into the Docker container, and then input the commands mentioned in this lesson in the terminal within the Docker container. A tutorial on how to enter the Docker container from the host can be found in this product's guide [00.Configuration and Operation Guide]-[3.Enter the Docker (For RPI5)]. For ORIN and RDK boards, you can directly open a terminal and enter the commands mentioned in this lesson.

3. Start the Program

- Enter the command in the terminal:

For Raspberry Pi 5 users, if the ROS container is not started, you need to start it first,

```
bringup_m3
```

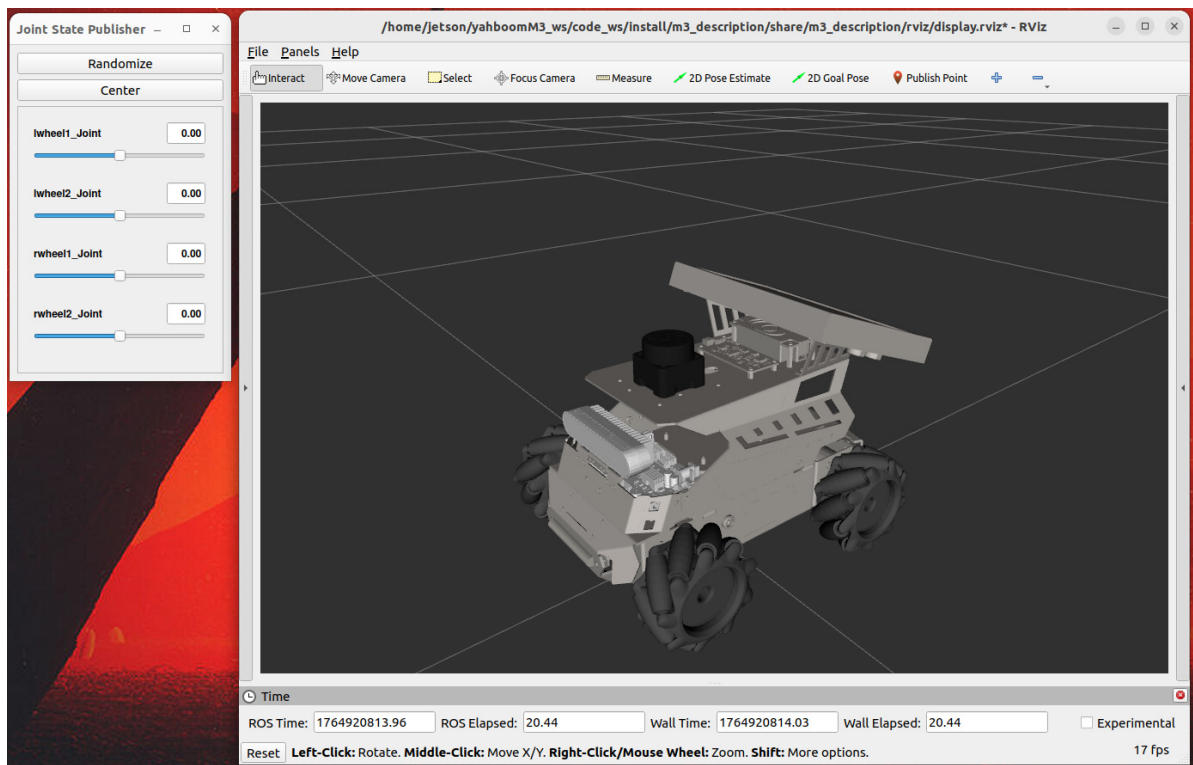
Open a terminal inside the container,

```
exec_m3
```

Enter the command,

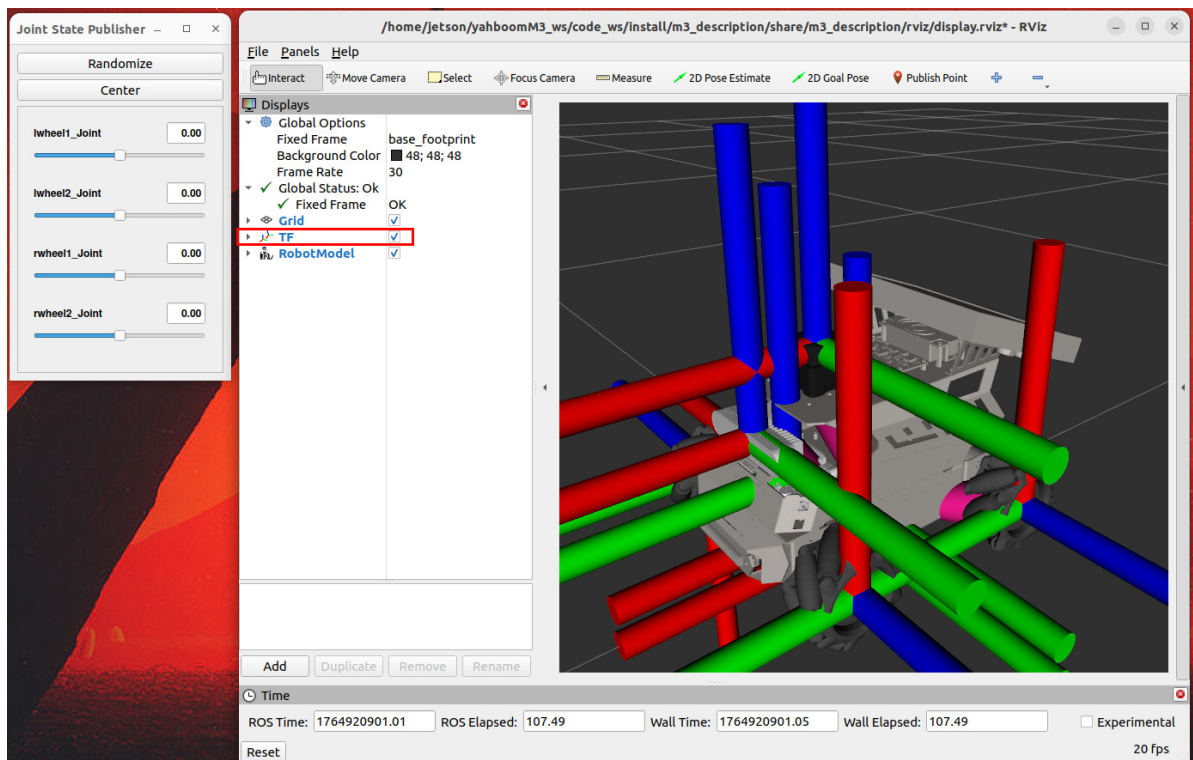
```
ros2 launch m3_description display.launch.py gui:=true
```

The startup interface is as follows:



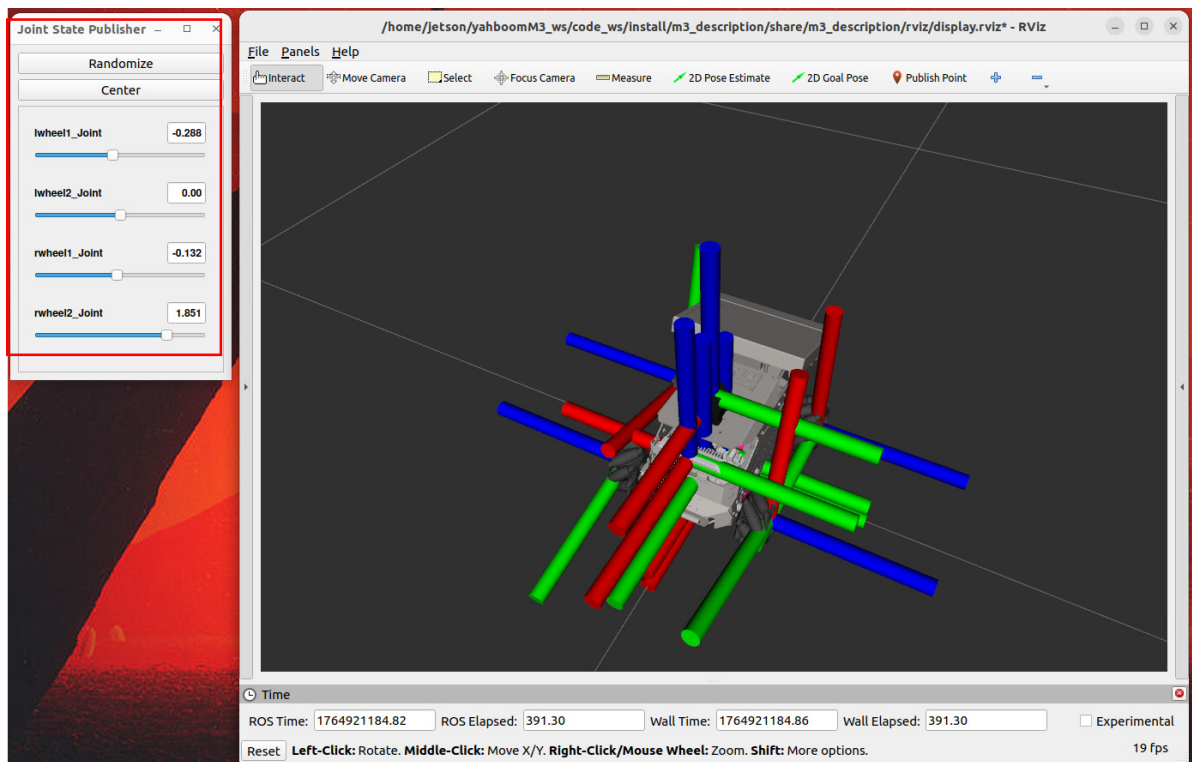
- Visualize the TF coordinate tree

Click on the left menu bar, then check the box for **TF** to visualize the TF coordinate tree.



- Test rotating joints

Click and drag the slider in the `joint_state_publisher` panel to adjust the rotation of the corresponding joint. For the ROSMASTER-M3 robot, the only movable joints are the four drive wheels.



4. Source Code Analysis

Robot model description file path:

```
~/yahboomM3_ws/code_ws/src/m3_description/urdf/ROSMaster-M3.xacro
```

Robot model launch file path:

```
~/yahboomM3_ws/code_ws/src/m3_description/launch/display.launch.py
```

- URDF, which stands for Unified Robot Description Format, is a robot model file described in XML format.

```
<?xml version="1.0" encoding="utf-8"?>

<robot name="ROSMaster-M3" xmlns:xacro="http://ros.org/wiki/xacro">

  <xacro:arg name="prefix" default="" />
  <!-- namespace -->
  <xacro:arg name="laser_type" default="dual" />
  <!-- Laser sensor type (single/dual) -->
  <xacro:arg name="camera_type" default="as_hp60c" />
  <!-- Depth camera type -->

  <xacro:include filename="$(find m3_description)/urdf/materials.xacro"/>
  <!-- Model color properties -->
  <xacro:include filename="$(find m3_description)/urdf/common_base.xacro" />
  <!-- Robot chassis template -->
  <xacro:include filename="$(find m3_description)/urdf/laser.xacro" />
  <!-- Lidar template -->
```

```

    <xacro:common_base prefix="$(arg prefix)" camera_type="$(arg camera_type)" />
<!-- Instantiating the robot chassis -->
    <xacro:laser laser_type="$(arg laser_type)" prefix="$(arg prefix)" />
<!-- Instantiate lidar -->

</robot>

```

The first line is a required XML item, describing the XML version information.

The second line describes the current robot's name; all information about the current robot is contained within the `<robot>` tag.

```

<!-- base_footprint projection point -->
<link
  name="{prefix}base_footprint">
</link>
<joint
  name="{prefix}base-joint"
  type="fixed">
  <origin
    xyz="0 0 0.03"
    rpy="0 0 0" />
  <parent
    link="{prefix}base_footprint" />
  <child
    link="{prefix}base_link" />
</joint>

<!-- base_link chassis -->
<link
  name="{prefix}base_link">
  <inertial>
    <origin
      xyz="-0.02984496350171 -0.000481153572635691 0.0765287728161752"
      rpy="0 0 0" />
    <mass
      value="0.99934444005899" />
    <inertia
      ixx="0.00160080186836152"
      ixy="3.98571758764143E-06"
      ixz="-0.000131328647411587"
      iyy="0.00221164246632588"
      iyz="5.5986256573897E-07"
      izz="0.00344837201415694" />
    </inertial>
  <visual>
    <origin
      xyz="0 0 0"
      rpy="0 0 0" />
    <geometry>
      <mesh
        filename="package://m3_description//meshes/base_link.STL" />
      </geometry>
    <material
      name="">

```

```

    <color
      rgba="0.776470588235294 0.756862745098039 0.737254901960784 1" />
    </material>
  </visual>
  <collision>
    <origin
      xyz="0 0 0"
      rpy="0 0 0" />
    <geometry>
      <mesh
        filename="package://m3_description//meshes/base_link.STL" />
      </geometry>
    </collision>
  </link>

```

4.1 Components

- **link**: Can be thought of as a person's arm.
- **joint**: Can be thought of as a person's elbow joint.
- Relationship between **link** and **joint**: Two links are connected by a joint. Imagine an arm with a forearm (**link**) and an upper arm (**link**) connected by an elbow joint (**joint**).
- In the URDF descriptive language, **link** is used to describe physical properties:
- Visual display is described by the **<visual>** tag.
- Collision properties are described by the **<collision>** tag.
- Physical inertia is described by the **<inertial>** tag (less commonly used).

Tag Descriptions:

- **origin**: Describes pose information; the **xyz** attribute describes the coordinate position in the larger environment, and the **rpy** attribute describes its own orientation.
- **mass**: Describes the mass of the link.
- **inertia**: The inertial reference frame. Due to the symmetry of the rotational inertia matrix, only the 6 upper triangular elements **ixx**, **ixy**, **ixz**, **iyy**, **iyz**, **izz** are needed as attributes.
- **geometry**: Describes the shape; the **mesh** attribute's main function is to load texture files, and the **filename** attribute specifies the path to the texture file.
- **material**: Describes the material; the **name** attribute is **required**, can be empty, and can be repeated. The **rgba** attribute within the **<color>** tag describes red, green, blue, and alpha (transparency), separated by spaces. The color range is [0-1].

4.2 Joints

Introduction

Describes the relationship between two joints, motion position and velocity limits, and kinematic and dynamic properties.

Joint Types:

- **fixed**: A fixed joint. Does not allow movement and serves as a connection.
- **continuous**: A rotational joint. Can rotate continuously with no angle limits.
- **revolute**: A rotational joint. Similar to **continuous**, but with angle limits.

- `prismatic`: A sliding joint. Moves along an axis, with position limits.
- `floating`: A floating joint. Has six degrees of freedom (3T3R).
- `planar`: A planar joint. Allows for translation or rotation on a plane.
- In the `<joint>` tag, the `name` attribute is **required**, describes the joint's name, and must be unique.
- In the `<joint>` tag, the `type` attribute corresponds to one of the six joint types.

Tag Descriptions

- **origin**: A sub-tag that refers to the relative position of the rotating joint in the parent's coordinate system.
- **parent, child**: Sub-tags that represent the two links to be connected; `parent` is the reference, and `child` rotates around the `parent`.
- **axis**: A sub-tag that indicates which axis (`xyz`) the `child` link rotates around and the amount of rotation around that fixed axis.
- **limit**: A sub-tag that primarily restricts the `child`. The `lower` and `upper` attributes limit the range of rotation in radians, the `effort` attribute limits the force range during rotation (positive/negative values, in Newtons or N), and the `velocity` attribute limits the speed of rotation (in meters/second or m/s).